Pathnex Training Module

Helm & Argo CD on Amazon EKS (Hands-on Lab)

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1. Introduction

This lab is designed for Pathnex students to learn:

- Helm → Application packaging tool for Kubernetes
- Argo CD → GitOps-based deployment tool
- Amazon EKS → Managed Kubernetes platform by AWS

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2. Prerequisites

Ensure the following are available:

- Running EKS cluster
- kubectl configured
- Helm installed
- AWS CLI configured

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3. Install Helm

```
curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
```

Verify:

```
helm version
```

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4. Connect to EKS Cluster

```
aws eks --region <REGION> update-kubeconfig --name <CLUSTER_NAME>
```

Verify:

```
kubectl get nodes
```

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5. Add Argo Helm Repository

```
helm repo add argo https://argoproj.github.io/argo-helm
helm repo update
```

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6. Create Namespace

```
kubectl create namespace argocd
```

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7. Install Argo CD using Helm (LoadBalancer Enabled)

```
helm install argocd argo/argo-cd \
  -n argocd \
  --set server.service.type=LoadBalancer
```

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8. Verify Installation

```
kubectl get pods -n argocd
```

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9. Get External LoadBalancer URL

```
kubectl get svc argocd-server -n argocd
```

Access UI:

```
https://<EXTERNAL-IP>
```

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10. Get Admin Password

```
kubectl get secret argocd-initial-admin-secret -n argocd \
 -o jsonpath="{.data.password}" | base64 -d
```

Login:

- Username: admin
- Password: (output above)

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11. Install Specific Helm Chart Version

```
helm search repo argo/argo-cd --versions
```

Install:

```
helm install argocd argo/argo-cd \
  -n argocd \
  --version <VERSION> \
  --set server.service.type=LoadBalancer
```

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12. Helm Release Management

List Releases

```
helm list -n argocd
```

View Release History

```
helm history argocd -n argocd
```

Rollback

```
helm rollback argocd <REVISION> -n argocd
```

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13. Argo CD Application Deployment (Pathnex App)

```
argocd app create pathnex-app \
  --repo https://github.com/argoproj/argocd-example-apps.git \
  --path guestbook \
  --dest-server https://kubernetes.default.svc \
  --dest-namespace default
```

Sync Application:

```
argocd app sync pathnex-app
```

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14. Verify Application

```
argocd app list
```

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15. Common Issues (EKS)

LoadBalancer Pending

Check subnet tagging:

```
kubernetes.io/role/elb=1
```

Pods Not Running

```
kubectl get pods -n argocd
```

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